

Linear algebra – 8

Block matrices and determinants of special shape

Homework

12 сентября 2026 года

Theoretical minimum. The Schur complement and the block determinant formula; $\det(E + AB) = \det(E + BA)$; Cauchy–Binet; circulants and the discrete Fourier transform; Vandermonde and Cauchy matrices; the Kronecker product with $\text{Spec}(A \otimes B) = \{\lambda_i \mu_j\}$ and $\det(A \otimes B) = (\det A)^m (\det B)^n$.

1. Prove the Cauchy–Binet formula and deduce that $\det(A^T A)$ equals the sum of squares of all maximal minors of A (the Gram identity), together with the volume interpretation.
2. (IMC) Define $A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_{n+1} = \begin{pmatrix} A_n & E_{2^n} \\ E_{2^n} & A_n \end{pmatrix}$. Prove that A_n has $n + 1$ distinct integer eigenvalues $\lambda_0 < \lambda_1 < \dots < \lambda_n$ with multiplicities $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.
3. (IMC) For each positive integer k find the smallest n_k for which there exist real $n_k \times n_k$ matrices $A_1, \dots, A_k$ with $A_1^2 = \dots = A_k^2 = O$, $A_i A_j = A_j A_i$ for all i, j , and $A_1 A_2 \dots A_k \neq O$. (*Hint: tensor products of 2×2 blocks.*)
4. Compute the Vandermonde determinant, and deduce both the Lagrange interpolation formula and the fact that n distinct points impose independent conditions on polynomials of degree $< n$.
5. Prove that the Cauchy matrix $(\frac{1}{x_i + y_j})$ has determinant $\frac{\prod_{i < j} (x_i - x_j)(y_i - y_j)}{\prod_{i, j} (x_i + y_j)}$, and deduce that the Hilbert matrix $(\frac{1}{i+j-1})$ is positive definite with a very small determinant.

6. (Romania national olympiad 2015) Let $m \geq n \geq 2$, let A be $m \times n$, B be $n \times m$, and let $a_0, \dots, a_k \in \mathbb{C}$ satisfy $a_0 E_m + a_1 AB + \dots + a_k (AB)^k = O_m$ while $a_0 E_n + a_1 BA + \dots + a_k (BA)^k \neq O_n$. Prove that $a_0 = 0$.
7. (Putnam 2023) For $i, j \in \{1, \dots, n\}$ let $s(i, j)$ be the number of pairs (a, b) of non-negative integers with $ai + bj = n$, and let S be the matrix with entries $s(i, j)$. Compute $\det S$.
8. (Cayley–Menger) Prove that the volume of a simplex with vertices $v_0, \dots, v_n$ is expressed through the determinant of the bordered matrix of squared distances, and deduce the condition for $n+2$ given numbers to be realisable as the distances of $n+2$ points in $\mathbb{R}^n$.
9. (IMC 2025 shortlist) Let D_n be the determinant of the $n \times n$ matrix whose diagonal entries are all 3, whose entries on the two neighbouring diagonals are all 1, and all of whose other entries are 0. Find a closed-form expression for D_n .
10. (Putnam 2024) Let $c_0, c_1, c_2, \dots$ be defined by $\frac{1-3x-\sqrt{1-14x+9x^2}}{4} = \sum_{k \geq 0} c_k x^k$ for small x , and for a positive integer n let A be the $n \times n$ matrix with (i, j) -entry c_{i+j-1} . Find $\det A$.